Distributions, Regression, and Inference — Comparing Models and Qualifying What They Support

Answer Key

Precalculus / Advanced Statistics

Quick Answers

1. 17.5	5. —	9. 128
2. 21.33	6. 22	10. 6
3. -4	7. 5	11. —
4. 0.5	8. $\frac{2}{3}$	12. —

Curriculum

Level: Precalculus / Advanced Statistics
HSS-IC.A–HSS-IC.B – *problems 5, 8, 11, 12*
HSS-ID.A–HSS-ID.C – *problems 1, 2, 3, 4, 6, 7, 9, 10*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

2. *If they got 17.5: reported the median as the mean, which a right-skewed distribution makes visibly different*
3. *If they got 4: computed predicted minus observed, which flips the sign and reverses what the residual says about the model*
6. *If they got 41: subtracted the smallest observation from the largest, reporting the range instead of the interquartile range*
8. *If they got 1: counted the two zero residuals as positive as well, so every hour looked like an under-prediction*

The argument this sheet builds. Summarise the data, measure each candidate model against it with residuals, then say how far the winner's evidence reaches. The final step is the one students skip: Model E fits Data Set A far better than Model L, and it *still* under-predicts at every hour it misses, on a run of six observations from a single experiment. A conclusion that names the better model without those two qualifications is not a stronger conclusion, only a less honest one.

1. Median of the observed counts.

The six values are already in increasing order, so the median is the mean of the third and fourth:

$$\frac{14+21}{2}$$

The median is $\boxed{17.5}$ hundred colonies.
With an even count there is no middle observation, so averaging the two middle ones is the definition, not a shortcut.

2. Mean of the observed counts, and what it says about shape.

$$6 + 9 + 14 + 21 + 31 + 47 = 128 \qquad \text{mean} = \frac{128}{6} = 21.333\ldots$$

To the nearest hundredth the mean is $\boxed{21.33}$ hundred colonies.
Mean against median: $21.33 > 17.5$. The mean sits above the median, which is the signature of a **right-skewed** distribution: the large values at hours 5 and 6 pull the mean up, while the median, which only cares about position, barely moves. That skew is itself a hint that growth here is multiplicative rather than additive — exactly the question problems 3 and 4 test directly.
Watch for: reporting 17.5 as the mean. On a skewed distribution the two summaries are visibly different, and which one you quote changes the story.

3. Model L's residual at $x = 3$.

Predict first, then subtract in the fixed order observed − predicted:

$$\text{predicted} = 8(3) - 6 = 18 \qquad \text{residual} = 14 - 18$$

The residual is $\boxed{-4}$.
What the sign means: negative means the observed count came in *below* the model, so Model L **over**-predicts at this hour — by 4 hundred colonies, which is more than a quarter of the observed value.
Watch for: computing predicted − observed, which returns +4 and reverses the verdict on the model.

4. Model E's residual at $x = 3$.

$$\text{predicted} = 4(1.5)^3 = 4(3.375) = 13.5 \qquad \text{residual} = 14 - 13.5$$

The residual is $\boxed{0.5}$.
Positive, so Model E under-predicts here — but by half a hundred colonies, one eighth of the size of Model L's miss at the same hour.

5. Which model fits better here, and why one residual cannot decide.

(a) **Model E fits better at that hour.** Compare the residuals by size, ignoring sign: $|0.5|$ against $|-4|$. Model E's miss is eight times smaller. Sign answers a different question (which direction the model errs); size answers this one.
(b) **Why one residual is not enough.** A model can land almost exactly on one point and miss badly everywhere else — a residual at a single x is one observation about the fit, not a summary of it. A defensible comparison examines:

- **all** the residuals, not one;
- their overall size, through the sum of squared residuals or the standard deviation of the residuals, so large misses are not cancelled by small ones;
- the **pattern** of their signs. Residuals scattered randomly above and below zero indicate the right shape of model; residuals that are all one sign, or that curve, indicate the wrong shape — no matter how small they are.

Problem 10 is exactly that check carried out on Model E.
Open explanation — $\boxed{\text{open response}}$; full credit needs the size comparison *and* at least two of the three items above.

6. Interquartile range of the observed counts.

Split the ordered six values into halves of three and take the median of each:

$$\text{lower half } 6,\ 9,\ 14 \Rightarrow Q_1 = 9 \qquad \text{upper half } 21,\ 31,\ 47 \Rightarrow Q_3 = 31$$

$$\text{IQR} = 31 - 9$$

The interquartile range is $\boxed{22}$ hundred colonies.
Why it differs from the range. The range here is $47-6 = 41$, and it is decided entirely by the two most extreme hours — on growing data, that means the last hour alone essentially sets it. The IQR describes the middle half of the run and is unmoved by either extreme, so on a right-skewed distribution it is the more stable description of typical spread.
Watch for: answering 41, which is the range, not the interquartile range.

7. Model L's residual at $x = 6$.

$$\text{predicted} = 8(6) - 6 = 42 \qquad \text{residual} = 47 - 42$$

The residual is $\boxed{5}$.
Compared with the earlier hour: at $x = 3$ Model L's residual was -4; here it is $+5$. The magnitude is similar but **the sign has flipped**. A linear model laid over data that curves upward must sit above the data in the middle and below it at both ends — so the sign flip is not noise, it is the fingerprint of using a straight line on curved data. Model L is not slightly off; it is the wrong shape.

8. Probability of drawing an hour with a positive residual.

The sample space is the six hours, all equally likely. The favourable outcomes are the four hours whose residual is strictly positive; a residual of exactly zero is neither positive nor negative and is not favourable.

$$P = \frac{4}{6}$$

In lowest terms, $P = \boxed{\frac{2}{3}}$.
Watch for: counting the two zero residuals as positive, giving $6/6 = 1$. That would say the model under-predicted at every single hour, which the residual table contradicts.

9. Total of the observed counts.

$$6 + 9 + 14 + 21 + 31 + 47$$

The six hours total $\boxed{128}$ hundred colonies.
Consistency check on problem 2: $128 \div 6 = 21.33$, the mean already found.

10. Where Model E's residual is largest, and what the pattern shows.

Scan the residual row for the largest entry: 1.4375, at hour $\boxed{6}$.

The pattern, in two parts.

- **Signs:** not one residual is negative. Four are positive and two are zero, so Model E never over-predicts — it sits on or below the data everywhere. Random scatter would give roughly as many negatives as positives, so this is a **systematic** bias, not noise.
- **Sizes:** they grow with the hour, from 0 to about 1.44. The model's error is not a fixed amount; it widens as the counts get larger, which is what happens when a model's growth rate is slightly too small.

What a bare "it fits well" hides: Model E's residuals are small, so any summary that only reports their size gives it a clean bill of health. The sign pattern says something a size summary cannot — the model is biased low, and extending it forward will under-predict by more, not less. That is precisely the qualification problems 11 and 12 demand.

11. What Model E can and cannot support about hour twelve.

What it supports. Inside the observed hours, Model E's residuals are small and its shape matches the multiplicative growth the data shows, so it is good evidence about the count at any time between hour one and hour six — including hours not measured, where the argument is interpolation.

What it does not support. Full credit needs all three of:

- **Extrapolation.** Hour twelve is twice as far out as any observation. Nothing in the data shows the pattern continuing there; the model would simply be assumed to hold.
- **The situation contradicts unbounded growth.** A real culture runs out of nutrients and space, so its curve levels off into an S-shape. Exponential growth is a good local description of the early phase and a bad global one. Model E at hour twelve returns a count far beyond anything the dish could hold.
- **One run is a sample of one.** Six measurements from a single experiment cannot separate what this culture did from what cultures do. Nothing here estimates run-to-run variability, so no margin of error can honestly be attached.

A good answer states the claim it *would* defend — the count within the observed range — rather than only listing objections.

Open explanation — $\boxed{\text{open response}}$.

12. Challenge: the conclusion paragraph.

A full-credit paragraph contains four things.

- **The choice, with evidence.** Model E fits better. At hour 3 its residual is 0.5 against Model L's -4; at hour 6 Model L is off by 5 while Model E is off by about 1.44. Model L's residuals also change sign across the run, the fingerprint of fitting a line to curved data, whereas Model E has the right shape.
- **The systematic weakness that survives.** Every one of Model E's residuals is zero or positive, so it under-predicts throughout, and the size of the miss grows with the hour. Its growth factor is a little too small. Being the better model does not make it unbiased.
- **A prediction with its range of defence.** A prediction anywhere from hour 1 to hour 6 is supported, with the note that Model E will read slightly low. Beyond hour 6 the claim is extrapolation, and the real culture will eventually level off, so no defence is offered there.
- **What a second run would add that arithmetic cannot.** Repeating the experiment gives an independent set of residuals. It shows whether the low bias is a property of the model or an accident of this culture, supplies run-to-run variability so a margin of error becomes meaningful, and tests whether the fitted growth factor reproduces. No further computation on these six numbers can produce any of that — more arithmetic on one sample is still one sample.

Open synthesis — $\boxed{\text{open response}}$; credit each of the four components separately.